

Models $\mathcal{M}$

Probabilistic models

- Standard distributions (e.g., Bern, Exp, Normal)
- Mixture models (e.g., GMM)
- Markov models, Hidden Markov models

Loss models

- MLP, DNN
- CNN
- RNN, LSTM

Choose model $M \in \mathcal{M}$ according to application.

- Parameters of model M : θ
- M is given by $p(\cdot \mid \theta)$
- Examples:
 - $\text{Bern}(\cdot \mid \theta), \mathcal{N}(\cdot \mid \mu, \Sigma)$
 - $p(\mathbf{x} \mid \theta) = \sum_{k=1}^K \pi[k] p(\mathbf{x} \mid \eta_k)$
 - $p(x_0, \dots, x_t \mid \pi, \mathbf{T}) = \pi(x_0) \mathbf{T}(x_0, x_1) \mathbf{T}(x_1, x_2) \cdots \mathbf{T}(x_{t-1}, x_t)$
 - HMM: $\pi, \mathbf{T}, \phi$
 - DNN: network weights θ

Infer model parameters θ from training set.

Find $\theta^* = \text{“optimal”}$ model parameter θ .

- MLE
- MAP
- EM algorithm
- Baum-Welch algorithm
- Sampling, MCMC
- Minimize loss or risk: stochastic gradient descent

Utilize the trained model.

Apply trained model $p(\cdot \mid \theta^*)$ in inference time.

- Goodness of fit
- Test accuracy, Precision, Recall, F1, AUC, etc.
- RMSE, MAE, etc.